



The effect of molybdenum on the corrosion behaviour of the high-entropy alloys $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ in aqueous environments¹

Y.L. Chou^{a,*}, J.W. Yeh^b, H.C. Shih^{a,b}²

^a Department of Materials Science and Engineering, National Tsing Hua University, Hsinchu 300, Taiwan³

^b Institute of Materials Science and Nanotechnology, Chinese Culture University, Taipei 111, Taiwan⁴

ARTICLE INFO⁵

Article history:

Received 21 September 2009⁶

Accepted 9 April 2010

Available online 13 May 2010

Keywords:

A. Alloy⁷

B. Polarisation

B. SEM

C. Pitting corrosion

ABSTRACT⁸

The purpose of this study is to investigate the electrochemical properties of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ high-entropy alloys in three aqueous environments which simulate acidic, marine, and basic environments at ambient temperature ($\sim 25^\circ\text{C}$). The potentiodynamic polarisation curves of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys, obtained in aqueous solutions of H_2SO_4 and NaOH , clearly revealed that the corrosion resistance of the Mo-free alloy was superior to that of the Mo-containing alloys. On the other hand, the lack of hysteresis in cyclic polarisation tests and SEM micrographs confirmed that the Mo-containing alloys are not susceptible to pitting corrosion in NaCl solution.

© 2010 Elsevier Ltd. All rights reserved.¹⁰

1. Introduction¹¹

In the past, the development of conventional alloy systems has been based primarily on one principal element, such as iron-, aluminium-, magnesium-, and nickel-based alloys, limiting the number of applicable alloy systems, even though a substantial amount of other elements is incorporated for property/processing enhancement [1,2]. In the 1990s, Inoue et al. [3] obtained the highly glass forming alloy by using at least three different elements with significantly different atomic radii. In addition, Greer [4] came up with the “confusion principle” in which he stated that the possibility of crystallisation could be lowered if the number of alloying elements was large. In the last two decades, a wide range of multicomponent amorphous alloys, including titanium-, nickel-, zirconium-, iron-, cobalt-, and copper-based alloy [5–10] systems have been developed. Multiple studies have been widely investigated the corrosion and passivation behaviour of these multicomponent amorphous alloys in a variety of electrolytes [11–34]. However, the design concept of afore-mentioned multicomponent amorphous alloys was, once again, based on one principal element. In an attempt to break through the cognitive realm of alloy design, a new multicomponent amorphous alloy system was subsequently developed [35–39]. This new alloy system is also called “high-entropy alloys” which are composed of multiprincipal elements in equimolar or near-equimolar ratios. Each element in the alloy sys-

tem is no more than 35 atomic percent (atom%). This study focuses on a new alloy design with multiprincipal elements in nonequimolar ratios.¹²

Solid solutions with multiprincipal elements tend to be more stable because they have larger mixing entropies. Following Boltzmann’s hypothesis of the relationship between entropy and system complexity, the configurational entropy change per mole, ΔS_{conf} , during the formation of a solid solution from six elements with equimolar fractions may be calculated from the following equation:¹⁴

$$\Delta S_{\text{conf}} = -k \ln \omega = -R \sum_{i=1}^6 x_i \ln x_i = R \ln 6 \approx 1.79R, \quad x_i = \frac{1}{6} \quad (1) \quad 15$$

where k is the Boltzmann’s constant, ω is the number of ways of mixing, and R is the gas constant, 8.314 J/mol/K. For instance, ΔS_{conf} of equimolar alloys with 3, 5, 7, and 9 elements are 1.10 R , 1.61 R , 1.95 R , and 2.20 R , respectively. In fact, when other positive vibrational, electronic, and magnetic contributions are considered, the entropy change associated with mixing in equimolar high-entropy alloy is greater than the calculated one [40]. Under the effects of high mixing entropies and the accompanied sluggish cooperative diffusion, they easily yield the formation of simple solid-solution phases and nanostructures during solidification, rather than many binary and ternary intermetallic compounds or other complicated phases [4,40]. The simplification of their structures avoids the disadvantages of difficulty in microstructure analyses and the inherent brittleness of ordered compounds.¹⁶

According to previous studies [35–39,41–45], these high-entropy alloy systems possess the following specialties: (1) ease of¹⁷

* Corresponding author. Address: Department of Materials Science and Engineering, National Tsing Hua University, 101, Section 2, Kuang-Fu Road, Hsinchu 300, Taiwan. Tel.: +886 3 5715131; fax: +886 3 5710290.

E-mail address: leng0122@yahoo.com.tw (Y.L. Chou).

amorphisation and nanoprecipitation in the alloy, (2) thermal stability, (3) high hardness [Vickers hardness (HV) = 700], (4) superior corrosion resistance, (5) excellent resistance to anneal softening, wear [1.2–1.8 m/mm³] and oxidation. Additionally, coating technology was further expanding their range of application, to include the hard-facing of golf heads and rollers [46], diffusion barrier [47], and soft magnetic films [48]. Among those previous studies, Reference 45 pointed out that Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x with various Mo contents (*x* values in the molar ratio, *x* = 0–1.0) is ductile, work hardenable, and strong at high temperature up to 800 °C, indicating that this alloy has potential application in high-temperature structures and working tools. However, the corrosion characteristics of the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloys are relatively unknown. Therefore, the purpose of this study is to investigate the electrochemical behaviour of the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloys under different environmental conditions.

2. Experimental 2

2.1. Test materials 3

Elements cobalt (Co), chromium (Cr), titanium (Ti), iron (Fe), and nickel (Ni) with purities of over 99 wt.% were present as granules as raw materials. The 39.5 wt.% Fe–60.5 wt.% Mo master alloy was used with the added molybdenum (Mo). This high-entropy alloy is a six-component alloy in the form of Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x, with 1.5 mol of Co, Ni, 0.5 mol of Ti, and 1 mol of the other two alloying elements and between zero and 1 mol of the added Mo. Table 1 presents the chemical compositions of the alloys in weight percent (wt.%). The multiprincipal-element Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloy system with various Mo contents with *x* from 0 to 1.0 was prepared in this work. An induction furnace was used to melt the high-entropy alloys at 1600 °C after purging with argon. First, raw materials of Co, Fe, and Ni were placed in the crucible. When Co, Fe, and Ni had melted in liquid state, then Ti, Cr, and FeMo were added to the crucible in order according to their melting points. Finally, the melt was poured into a shell mold preheated at 1100 °C. The size of the ingots was 8 × 7 × 14 (cm). The high-entropy alloy cylinder for electrochemical measurements was obtained by cutting the bulk material by electric arc line into pieces that were 3 mm thick and 8 mm in diameter. Each test specimen was then cold mounted, using an epoxy resin to give an exposed surface area of 0.5 cm² for the high-entropy alloys. Before each electrochemical experiment, all specimens were mechanically ground using a series of 240–600 silicon carbide (SiC) grit papers and cleaned with acetone and distilled water.

2.2. Structure analysis of the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloy 5

The microstructure and composition analyses of the as-cast alloys were carried out by JEOL 5410 scanning electron microscope (SEM) equipped with an energy dispersive X-ray spectrometry (EDS). X-ray diffraction (XRD) using Cu Kα radiation was used to identify crystalline phases and obtain the lattice parameters of these alloys. The XRD investigation was performed with the diffraction angle (2θ) from 20° to 100° at a speed of 4°/min.

Table 1
Chemical composition (wt.%) of the test materials. 12

Designation	Composition (atomic mole ratio)	Co _(58.93) (%)	Cr _(52.00) (%)	Fe _(55.85) (%)	Ni _(58.69) (%)	Ti _(47.88) (%)	Mo _(95.94) (%)
Mo ₀	Co _{1.5} CrFeNi _{1.5} Ti _{0.5} Mo ₀	28.68	16.87	18.12	28.56	7.77	0
Mo _{0.1}	Co _{1.5} CrFeNi _{1.5} Ti _{0.5} Mo _{0.1}	27.81	16.36	17.57	27.70	7.53	3.02
Mo _{0.5}	Co _{1.5} CrFeNi _{1.5} Ti _{0.5} Mo _{0.5}	24.82	14.60	15.68	24.72	6.72	13.47
Mo _{0.8}	Co _{1.5} CrFeNi _{1.5} Ti _{0.5} Mo _{0.8}	22.96	13.51	14.51	22.87	6.22	19.94

2.3. Electrochemical measurements 7

The corrosion resistance of Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloys was conducted in three different environmental conditions; 1 molar concentration (M) sodium hydroxide (NaOH), 0.5 M sulfuric acid (H₂SO₄), and 1 M sodium chloride (NaCl), which will be referred to as base, acid, and salt solution, respectively henceforth at 25 °C under atmospheric pressure. The test solution was deaerated by bubbling purified nitrogen gas before and throughout the electrochemical tests to eliminate the effect of dissolved oxygen. The electrochemical polarisation was performed in a typical three electrode cell that consisted of a specimen as the working electrode, a silver/silver chloride (Ag/AgCl) in 3 M potassium chloride (KCl) as the reference electrode, and a platinum (Pt) sheet as the counter electrode. The Ag/AgCl scale (3 M KCl) is 0.208 V lower than the standard hydrogen electrode (SHE) [49]:

$$E, V_{\text{Ag/AgCl}} + 0.208 = E, V_{\text{SHE}} \quad 9$$

Unless otherwise stated in this paper, all potentials recorded were referred to standard hydrogen electrode. 10

Potentiodynamic polarisation measurements were made at a scan rate of 1 mV/s from an initial potential of −0.5 V to a final potential of 1.5 V versus the open current potential (OCP). The potential was controlled and the current was measured using a potentiostat (Autolab PGSTAT30). A cyclic polarisation technique started at −0.5 V by scanning at 1 mV/s towards more positive potentials up to 1.5 V versus the OCP, at which value the potential scan was reversed and returned to where the polarisation began. Before either of these tests was conducted, the OCP was recorded for 15 min to yield a steady-state potential. Then, the specimen was cathodically polarised to a potential of −192 mV_{SHE} for 5 min to reduce the existing surface films. Following electrochemical testing, multiple electrochemical parameters were determined: *E*_{corr}, corrosion potential, i.e., the potential where zero current density [21]; *E*_{pp}, primary passivation potential, the potential at which the maximum anodic current density reached in the active region [50]; *E*_{sp}, secondary passivation potential, the potential at which the maximum anodic current density reached in the passive region [51]; *E*_b, the potential at which the current density suddenly increases is determined to be the breakdown potential [20]; Δ*E*, the difference between *E*_b and *E*_{pp} is the estimated passive region width [42]; *E*_{prot}, the protection potential where the forward and reverse scan crosses in the passive region [51]; *E*_{rp}, the repassivation potential is considered to be the potential at which the cyclic polarisation curves are approximately horizontal on the potential reverse scan [42]; *i*_{crit}, critical current density, the maximum anodic current density in the active region [50]; *i*_{pass}, anodic current density values remain completely constant in the passive range [50]; the corrosion current density, *i*_{corr}, is obtained from the curves using the intersection of the cathodic linear extrapolation (approximately −50 to −100 mV from the corrosion potential) and the corrosion potential horizontal line. 11

8

10

11

13

2.4. Surface morphology and chemical analysis 1

Following the polarisation experiment, the specimen was cleaned using distilled water and then dried in nitrogen. Immediately thereafter, the morphology and chemical composition of the corroded surface of the high-entropy alloy was investigated using a JEOL 5410 SEM.

3. Results and discussion 3

3.1. Microstructure of the as-cast $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys 4

Fig. 1 illustrates the XRD patterns of the as-cast $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0, 0.1, 0.5, 0.8$) alloys. In this study, all these alloys basically belong to a simple FCC solid-solution structure. However, as Mo content in $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys increases up to 0.5 and 0.8 mol, the small peaks slightly on the left and right of the matrix FCC in XRD patterns are identified as Cr and Mo rich sigma (σ) phase. As the Mo content decreases in these alloys, the peak intensities also change due to the lattice distortion caused by the different atomic size and still remains in FCC structure [45]. There is also no tendency to form complex or multiple phases when increasing the number of constituent elements. Because the wavelength of X-ray is constant ($\lambda = 1.54 \text{ \AA}$), from the angle (θ) and plane (hkl) of each diffraction peak, a set of lattice parameter can be calculated by Bragg's law. These values of lattice parameter were plotted against $\cos^2 \theta$, and the precise parameter of alloy (a_0) can be estimated by least squares and extrapolation method [52]. The lattice parameters of $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0, 0.1, 0.5, 0.8$) alloys in this study are about $3.59 \text{ \AA}$. Fig. 2 shows the SEM images of the as-cast $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys, which are composed of dendrites and interdendrites. According to the EDS analyses as listed in Table 2, the interdendrite (lighter phase) of the as-cast $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0, 0.1$) alloy is Ni and Ti rich phase, and dendrite (darker phase) is Fe and Cr rich phase. Table 3 lists chemical mixing enthalpy (ΔH^{chem}) of a pair of atoms [53–55]. As can be seen in Table 3, Ni and Ti have the larger negative enthalpy in the alloy. Thus, they are likely to aggregated in the interdendrite phase. When increasing the Mo content in the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0.5, 0.8$) alloys, the interdendrites are rich in Cr and Mo, while the dendrites are rich in Co and Ni.

As listed in Table 3, Mo element has weaker binding force to other constituent elements than Fe, Co, Ni, Ti, and Cr. However, the chemical analysis of that “interdendrites” is presented in

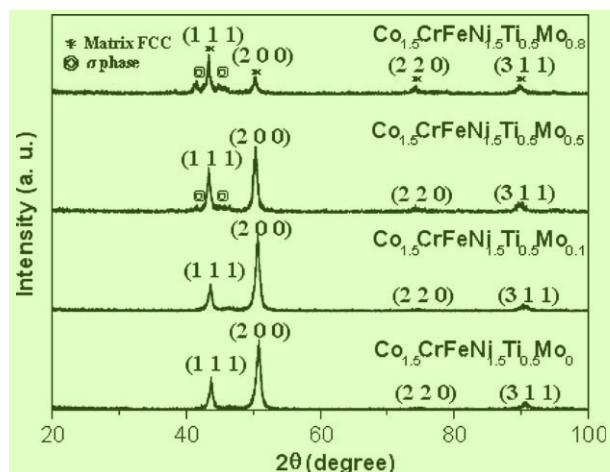


Fig. 1. XRD patterns of the as-cast $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0, 0.1, 0.5, 0.8$) high-entropy alloys.

Fig. 2c, d and Table 2, displaying Cr and Mo rich σ phase, as compared to the whole $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0.5, 0.8$) alloys composition (Table 1). The tendency to σ phase formation of an austenitic stainless steel can be known from the equivalent chromium content (ECC) equation [56]:

$$\begin{aligned} \text{ECC} = \% \text{ chromium (Cr)} &+ 0.31\% \text{ manganese (Mn)} \\ &+ 1.76\% \text{ molybdenum (Mo)} + 0.97\% \text{ tungsten (W)} \\ &+ 2.02\% \text{ vanadium (V)} + 1.58\% \text{ silicon (Si)} \\ &+ 2.44\% \text{ titanium (Ti)} + 1.7\% \text{ niobium (Nb)} \\ &+ 1.22\% \text{ tantalum (Ta)} - 0.266\% \text{ nickel (Ni)} \\ &- 0.177\% \text{ cobalt (Co)} \end{aligned} \quad (3)$$

If the ECC is greater than 17–18%, the steel is susceptible to σ phase formation. In this case, according to the ECC equation mentioned above, the ECC values for the $\text{Co}_{1.5}\text{CrFeNi}_{0.5}\text{Ti}_{0.5}\text{Mo}_{0.5}$ and $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.8}$ alloys were 43.73% and 53.63%, respectively. This indicates that an ECC of above 43% is susceptible to form that phase. Moreover, according to Conejero et al. [57], the Cr and Mo rich σ phase forms mainly at grain boundaries in 316L type stainless steels and full austenitic alloys during the solidification.

3.2. Potentiodynamic polarisation curves 12

Fig. 3 plots the potentiodynamic polarisation curves of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys with various Mo contents ($x = 0, 0.1, 0.5$, and 0.8) in $0.5 \text{ M H}_2\text{SO}_4$ solution. The potentiodynamic polarisation curve exhibited active–passive corrosion behaviour, and the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys yielded a wide passive region ($E_b - E_{pp} = \Delta E > 1100 \text{ mV}$), which indicated a tendency of the alloys to passivate. Moreover, the passive region of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_0$ alloy was broader than that of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.1}$, $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.5}$, and $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.8}$ alloys. The E_{sp} occurred for the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.8}$ alloy under $0.5 \text{ M H}_2\text{SO}_4$ solution, which may be attributed to the formation and dissolution of the molybdenum oxide film. Table 4 presents the electrochemical parameters of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0, 0.1, 0.5$, and 0.8) in deaerated $0.5 \text{ M H}_2\text{SO}_4$ solution. The corrosion potential (E_{corr}) of the Mo-containing alloys was more apparently more noble than that of the Mo-free alloy. However, the corrosion current densities (i_{corr}) of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.1}$, $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.5}$, and $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.8}$ alloys were 78, 72, and $69 \mu\text{A}/\text{cm}^2$, respectively; the i_{corr} value dropped to $30 \mu\text{A}/\text{cm}^2$ when Mo was eliminated from the alloy, as in $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_0$. Furthermore, the passive current was increased in those alloys with Mo content. The primary passivation potential (E_{pp}) of the Mo-free alloy was lower than that of the Mo-containing alloys. The critical current density (i_{cr}) of the Mo-free alloy was a little higher than that of the Mo-containing alloys. Hence, the Mo alloying element addition has negative effect on the corrosion resistance of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys.

The broadest passive region and lowest passive current density under $0.5 \text{ M H}_2\text{SO}_4$ solution (Table 4) were registered in the Mo-free alloy whose main characteristic is the highest presence of Cr [58]. Although the highest Mo content of $\text{Co}_{1.5}\text{CrFeNi}_{0.5}\text{Ti}_{1.5}\text{Mo}_{0.8}$ alloy, lower Cr content with respect to $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_0$ alloy, generated a higher passive current density along the passive region. In the passive region, the measured current comes from the dissolution of the oxide film. The dissolution current of the oxide film increases as the Cr content in the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0, 0.1, 0.5$, and 0.8) alloys decreases, indicating that the addition of Cr can improve the stability of the oxide film. The increase in passive current density with decreasing Cr content is either due to a change in structure or a decrease in thickness of the passive

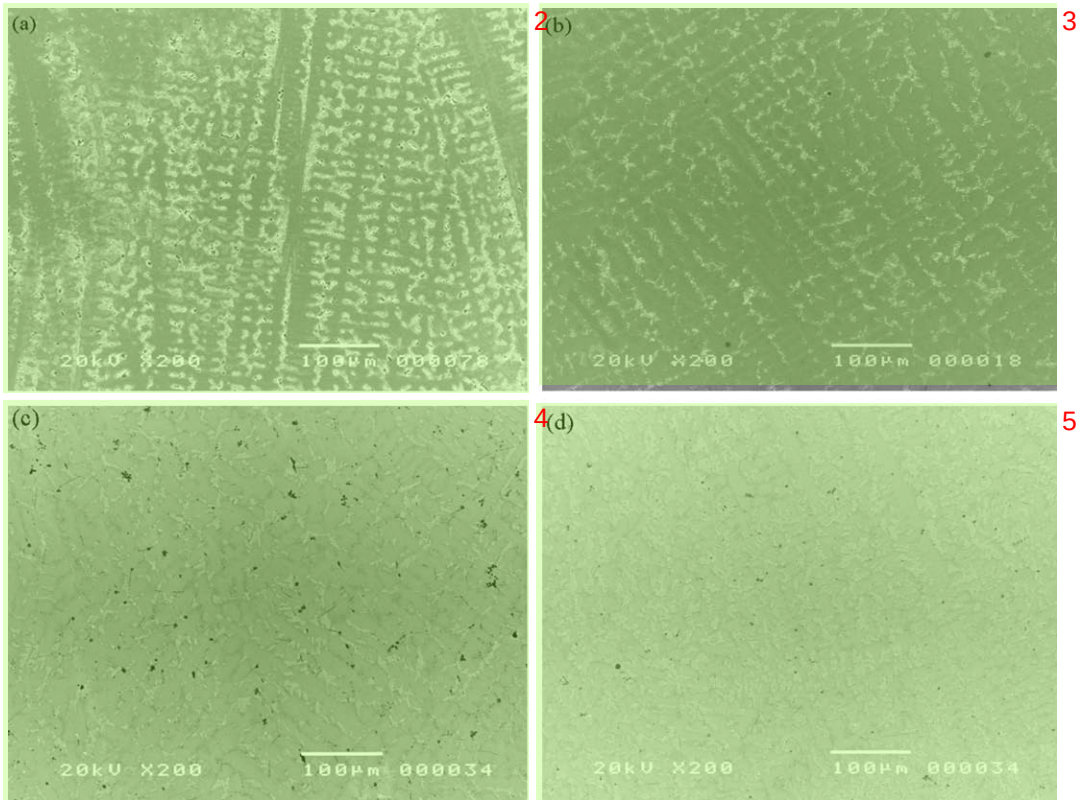


Fig. 2. SEM images of the as-cast $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ high-entropy alloys: (a) $x = 0$, (b) $x = 0.1$, (c) $x = 0.5$, and (d) $x = 0.8$ mol.

Table 2
EDS analyses of the dendrite and interdendrite structure of the as-cast.

Element	$\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_0$ (atom%)		$\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.1}$ (atom%)	
	Interdendrite	Dendrite	Dendrite	Dendrite
<i>$\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ high-entropy alloys</i>				
Co	26.7	27.6	25.8	26.6
Cr	15.1	21.0	13.3	19.1
Fe	15.3	19.8	14.1	18.3
Ni	28.9	25.9	29.5	26.8
Ti	14.0	5.7	15.5	7.3
Mo	–	–	1.8	1.9
	$\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.5}$ (atom%)		$\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.8}$ (atom%)	
	Interdendrite	Dendrite	Dendrite	Dendrite
Co	21.0	24.0	20.6	24.8
Cr	20.5	18.1	19.6	15.5
Fe	14.2	16.9	15.2	16.6
Ni	13.0	26.2	14.0	28.2
Ti	7.1	7.5	4.3	7.3
Mo	24.2	7.3	26.3	7.6

film. It is generally accepted that low solubility and high stability of chromic oxide (Cr_2O_3) based products reduce the passive current density and increase the size of the passive region on the pH-potential diagram [59].

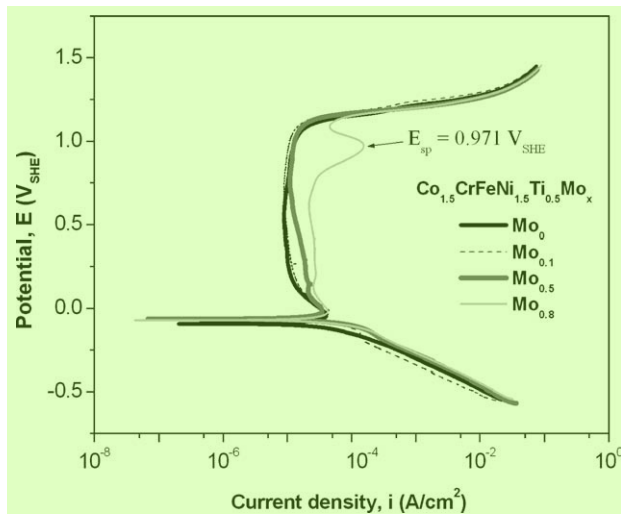
The addition of Mo is widely recognised for its beneficial effect on the corrosion resistance of the stainless steels and has been studied extensively in the literature [60–65]. Sugimoto and Sawada [60] concluded that Mo addition to austenitic stainless steels allows formation of a passive film consisted of a Cr (III) oxyhydroxide containing Mo^{6+} ions, which reduces the corrosion current density in the active region for acidic solution. However, Hashimoto et al. [61] proposed that the main effect of Mo was to decrease the rate of dissolution in active zones by formation and retention

of Mo (VI) oxyhydroxide or molybdate (MoO_4^{2-}) at these sites. Furthermore, Ogawa et al. [62] suggested that Mo is first dissolved from the substrate into the solution and then oxidised to MoO_4^{2-} .

As mentioned above, many authors conclude that molybdenum species, present in the passive layer, are mainly ionic compounds (MoO_4^{2-}), which are adsorbed on the top of the material surface, acting as a barrier against the electrochemical attack [63–67]. However, given the 0.5 M H_2SO_4 acidic conditions ($\text{pH} < 1$) in this study and according to Pourbaix diagrams [68], an influence of Mo via MoO_4^{2-} formation should be considered improbable, since, in these conditions, MoO_4^{2-} is unstable thermodynamically and tends to precipitate as Mo (VI) oxide, e.g., MoO_3 . Hence, these dia-

Table 3Chemical mixing enthalpy (ΔH^{chem}) of a pair of atoms [53–55].

System	ΔH^{chem} (kJ/mol)
Mo–Co	–5
Mo–Cr	0
Mo–Fe	–2
Mo–Ni	–7
Mo–Ti	–4
Fe–Co	–1
Fe–Cr	–1
Fe–Ni	–2
Fe–Ti	–9
Co–Ni	0
Co–Ti	–28
Ni–Ti	–35

**Fig. 3.** Potentiodynamic polarisation curves of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys with various Mo contents ($x = 0, x = 0.1, x = 0.5$, and $x = 0.8$ mol) in deaerated 0.5 M H_2SO_4 .

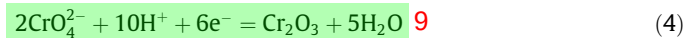
grams establish MoO_3 as the only compound formed in 0.5 M H_2SO_4 .

The obtained polarisation results (Table 4) have revealed that Mo addition tends to increase the corrosion current density of Mo-containing alloys in 0.5 M H_2SO_4 . These results are in disagreement with Hashimoto [69] who reported that in the active region of stainless steels, where preferential dissolution of iron and chromium occurred, the formation of the passive molybdenum oxide film was responsible for a large decrease of the current density. The opposite results were found by Bojinov et al. [70] in studies of the electrochemical behaviour of anodic films on pure Cr,

Fe–Cr alloys and Fe–Cr–Mo alloys in 1 M H_2SO_4 solution. The corrosion current density in the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.1}$ alloy is lower than that in the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_0$ alloy, which may be attributed to the simultaneous presence of lower Cr and slightly higher Mo content (1.8 atom%) in the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.1}$ alloy (Table 2). As the Mo content increases from 0.5 to 0.8 mol in the high-entropy alloys, the interdendrite phase of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys is a Cr and Mo rich σ phase, and the dendrite is a Cr and Mo depleted phase. The σ phase is probably the most studied and undesirable phase in a stainless steel. Formation of σ phase can occur in the austenitic, ferritic, and duplex stainless steels. Especially in duplex stainless steels, the formation of σ phase is accelerated by increased amount of Cr and Mo [71,72]. The formation of that phase not only causes losses in ductility and toughness of the steel, but also reduces its corrosion resistance by removing Cr and Mo from the austenitic matrix [73]. Conejero et al. [57] studied premature corrosion failure of a 316L stainless steel plate exposed to a H_2SO_4 environment. It was reported that a 316L stainless steel plate has failed due to the presence of σ phase, which is deleterious to corrosion performance of the stainless steels in H_2SO_4 solution. Therefore, the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0.5, 0.8$) alloys possessed the lower corrosion current density in comparison with the Mo-free alloy in this study, which could be also attributed to the presence of σ phase.

Fig. 4 shows the potentiodynamic polarisation curves of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys with various Mo contents ($x = 0, 0.1, 0.5$, and 0.8) in 1 M NaOH solution. The corresponding E_{pp} varied from -307 to -545 mV_{SHE} and the E_{sp} was observed between 317 and 335 mV_{SHE} for the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0, 0.1, 0.5$, and 0.8) alloys under 1 M NaOH solution. Table 5 lists the electrochemical parameters of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0, 0.1, 0.5$, and 0.8) in deaerated 1 M NaOH solution. The i_{cri} for the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0, 0.1, 0.5$, and 0.8) alloys increased with Mo contents, implying that the barrier to passivation increased with the increase of Mo content. The i_{corr} for the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0, 0.1, 0.5$, and 0.8) alloys increased with Mo contents and the E_{corr} varied from -379 to -657 mV_{SHE}. Thus, adding Mo to the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys reduces the corrosion resistance under 1 M NaOH solution.

Stainless steels and alloys containing Cr and Mo are susceptible to transpassive dissolution, because the oxides of these elements are unstable at high anodic potentials. Cr dissolution can take place due to the oxidation of solid Cr (III) oxide to soluble Cr (VI); similarly, Mo (IV) oxide is oxidised to soluble Mo (VI). According to the Pourbaix diagrams [68], the relevant equilibria are



$$E_0, V_{\text{SHE}} = 1.311 - 0.0985 \text{ pH} + 0.0197 \log(\text{CrO}_4^{2-}) \quad (4') \quad (12)$$

Table 4Electrochemical parameters of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys in deaerated 0.5 M H_2SO_4 .

Designation	E_{corr}^a (mV _{SHE})	i_{corr}^b ($\mu\text{A}/\text{cm}^2$)	E_{pp}^c (mV _{SHE})	i_{cri}^d ($\mu\text{A}/\text{cm}^2$)	i_{pass}^e ($\mu\text{A}/\text{cm}^2$)	E_{sp}^f (mV _{SHE})	E_b^g (mV _{SHE})	ΔE^h (mV)
Mo ₀	–92	30	–41	46	9	–	1089	1130
Mo _{0.1}	–71	78	–23	45	9	–	1089	1112
Mo _{0.5}	–64	72	–20	41	11	–	1089	1109
Mo _{0.8}	–70	69	–20	44	22	971	1089	1109

^a E_{corr} : corrosion potential.^b i_{corr} : corrosion current density.^c E_{pp} : primary passivation potential.^d i_{cri} : critical current density.^e i_{pass} : passive current density.^f E_{sp} : secondary passivation potential.^g E_b : breakdown potential.^h ΔE : a potential difference between E_b and E_{pp} .

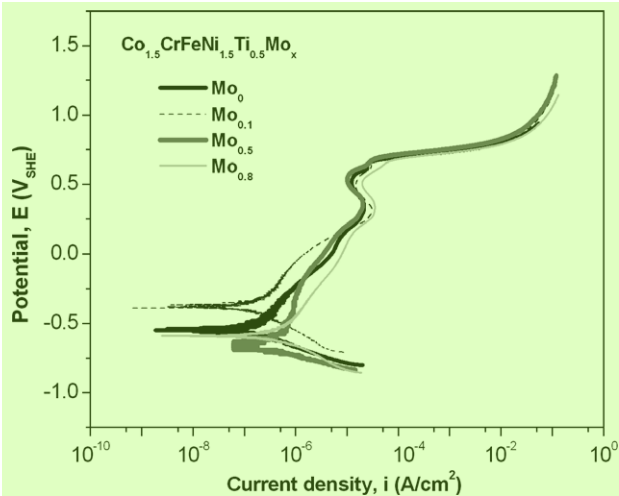
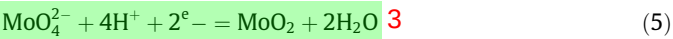


Fig. 4. Potentiodynamic polarisation curves of the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloys with various Mo contents ($x = 0$, $x = 0.1$, $x = 0.5$, and $x = 0.8$ mol) in deaerated 1 M NaOH.



$$E_0, V_{\text{SHE}} = 0.606 - 0.1182 \text{ pH} + 0.0295 \log(\text{MoO}_4^{2-}) \tag{5'}$$

It can be observed that Mo transpassivity may occur at a significant lower potential than that of Cr. The product of the transpassive region of Mo depends on the pH of test solutions. Therefore, MoO₄^{2−} is formed in NaOH solutions in this study.

The Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x ($x = 0, 0.1, 0.5$, and 0.8) alloys achieve their corrosion resistance by formation of an oxide passive layer. It has been reported that Ni, Ti, Co, and their alloys resist corrosion by NaOH solutions [74]. For Ni and its alloys, the generally accepted Ni (II) oxide passive film contributes to retard the corrosion phenomenon by diminishing the corrosion current density (i_{corr}) although it is not the essential element to improve the passive properties of the alloys. The Ti and its alloys have good protective properties due to its compact, passive film that contains the protective titanium dioxide (TiO₂), corrosion and passivation behaviour of Co and its alloys in aqueous NaOH solutions are covered by a stable passive film which consists of Co (II) oxide/hydroxide. Therefore, from Table 5 it is clear that the i_{corr} and i_{cr} for the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x ($x = 0, 0.1, 0.5$, and 0.8) alloys increased with the increase of Mo contents in 1 M NaOH solutions, which could also be attributed to the simultaneous presence of low Ni, Ti, and Co content in the Mo-containing alloys in comparison with the Mo-free alloy.

3.3. Cyclic potentiodynamic polarisation

A cyclic polarisation technique was adopted to determine whether the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloys ($x = 0, 0.1, 0.5$, and 0.8)

suffered from pitting corrosion in 1 M NaCl solution at approximately 25 °C. The hysteresis revealed by the cyclic polarisation curve provides information on the pitting corrosion rate and how readily a passive film repairs itself. Negative hysteresis occurs when the reverse scan current density is less than that for the forward scan, and positive hysteresis occurs when the reverse scan current density exceeds the forward scan current density [75].

Cyclic polarisation measurements were made at a scanning rate of 1 mV/s. The potential scan began at −0.5 V and continued in the anodic direction until the potential was 1.5 V versus the OCP, at which value the potential scan was reversed and returned to where the polarisation began. Fig. 5 plots the cyclic polarisation curves of the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloys with various Mo contents ($x = 0, 0.1, 0.5$, and 0.8). The black and gray arrows next to the forward and reverse anodic branches indicate the potential scan directions. In terms of the electrochemical parameters shown in Table 6, the addition of Mo moves the E_b in the noble direction. The cyclic polarisation curve of the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo₀ alloy was referred to as a pseudo-passive curve because the curve did not exhibit a distinct E_{pp} , but the E_b was observed. The difference between E_b and E_{pp} was the estimated passive region width ΔE . The ΔE for the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo₀ alloy was 557 mV. As the Mo content in the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloys increased from 0.1 to 0.8 mol, the ΔE varied between 1371 and 1429 mV. Therefore, the ΔE of the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloys broadened due to the effect of Mo, which is in agreement with Sedriks et al. [58]. The Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloys ($x = 0.1, 0.5$, and 0.8) alloys showed negative hysteresis in the cyclic polarisation curve, indicating that the Mo-containing alloys were resistant to pitting corrosion in 1 M NaCl solution, as can be seen in Fig. 5b–d. However, the positive hysteresis of the cyclic polarisation curve demonstrates that pitting of the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo₀ alloy can be induced in 1 M NaCl solution, as in Fig. 5(a). The area contained within a positive hysteresis loop is related to the amount of pit propagation that occurs during the cycle, according to Wilde [76]. The intersection at the passive region defined protection potential (E_{prot}), below which the established pits on metal surface apparently could not keep on growing. In contrast, new pits would initiate only above E_b . Between E_{prot} and E_b , new pits could not initiate, but old ones still could grow. Furthermore, the i_{corr} of the Mo-containing alloys was lower than that of the Mo-free alloy. Therefore, adding Mo element has positive effect on the corrosion resistance of Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloys in 1 M NaCl solution.

It has been well established that the addition of Mo to stainless steels improves the pitting corrosion resistance in media containing chloride [50,58,77–78]. It has been considered that Mo could form MoO₄^{2−} ions in the solid state in the outer regions of the film in neutral NaCl solutions according to Pourbaix diagrams [68]. This MoO₄^{2−} layer is cation selective and can serve as a very effective pitting inhibitor, which resists the incorporation of anions such as Cl[−] and Br[−], allowing for the growth of the Cr oxide inner barrier layer. The resulting bipolar layer stabilises the oxide phase [65].

Table 5 Electrochemical parameters of the Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x alloys in deaerated 1 M NaOH.

Designation	E_{corr}^a (mV _{SHE})	i_{corr}^b (μA/cm ²)	E_{pp}^c (mV _{SHE})	i_{cr}^d (μA/cm ²)	E_{sp}^e (mV _{SHE})	E_b^f (mV _{SHE})
Mo ₀	−546	0.09	−485	0.20	335	653
Mo _{0.1}	−379	0.11	−307	0.23	317	653
Mo _{0.5}	−657	0.16	−545	0.62	328	653
Mo _{0.8}	−589	0.25	−509	0.63	317	660

^a E_{corr} : corrosion potential.
^b i_{corr} : corrosion current density.
^c E_{pp} : primary passivation potential.
^d i_{cr} : critical current density.
^e E_{sp} : secondary passivation potential.
^f E_b : breakdown potential.

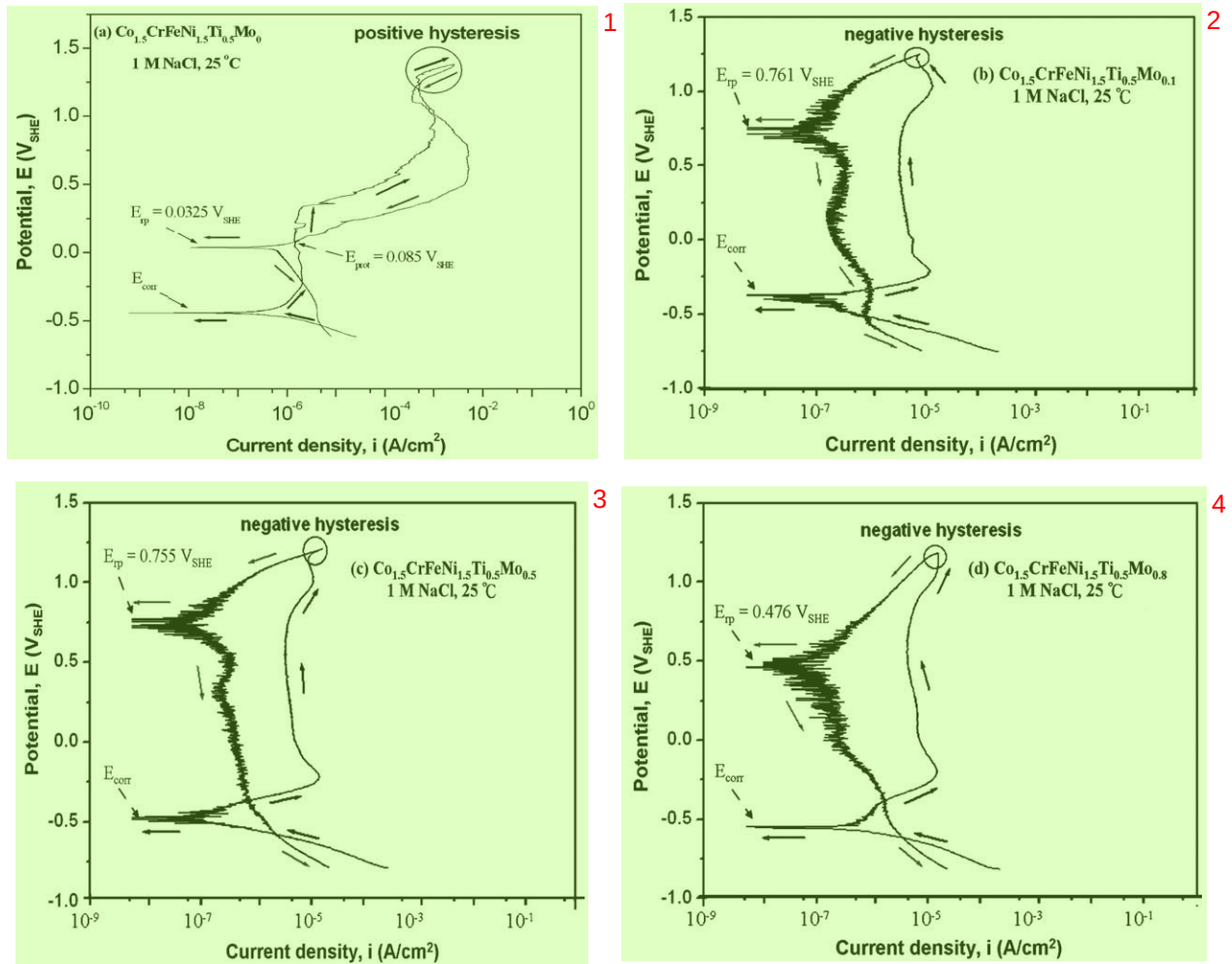


Fig. 5. Cyclic polarisation curves of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys with various Mo contents (a) $x = 0$, (b) $x = 0.1$, (c) $x = 0.5$, and (d) $x = 0.8$ mol in deaerated 1 M NaCl.

Table 6

Electrochemical parameters of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys in deaerated 1 M NaCl.

Designation	E_{corr}^a (mV _{SHE})	i_{corr}^b ($\mu\text{A}/\text{cm}^2$)	E_{pp}^c (mV _{SHE})	i_{pass}^d ($\mu\text{A}/\text{cm}^2$)	E_b^e (mV _{SHE})	E_{prot}^f (mV _{SHE})	E_{rp}^g (mV _{SHE})	ΔE^h (mV)
Mo_0	−443	0.57	−219	1.46	338	85	32.5	557
$\text{Mo}_{0.1}$	−381	0.13	−215	3.80	1214	–	761	1429
$\text{Mo}_{0.5}$	−493	0.20	−219	4.10	1164	–	755	1383
$\text{Mo}_{0.8}$	−551	0.41	−189	5.11	1182	–	476	1371

^a E_{corr} : corrosion potential.

^b i_{corr} : corrosion current density.

^c E_{pp} : primary passivation potential.

^d i_{pass} : passive current density.

^e E_b : breakdown potential.

^f ΔE : a potential difference between E_b and E_{pp} .

^g E_{prot} : protection potential.

^h E_{rp} : repassivation potential.

It has been reported that the content of Cr and Mo plays an important role in the corrosion resistance of stainless steels and nickel alloys [79–81]. For Fe–Ni–Cr–Mo stainless steel, such as 316 stainless steels, the relative effectiveness of Cr and Mo contents on pitting or crevice corrosion usually can be assessed qualitatively by pitting resistance equivalent (PRE), which is represented by the empirical equation [82]:

$$\text{PRE} = \% \text{Cr} + 3.3\% \text{Mo} \quad (6)$$

In this study, according to the PRE equation mentioned above, the PRE values for the pitting-resistant $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.1}$,

$\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.5}$, and $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.8}$ alloys are 26.32, 59.05, and 79.31, respectively. The $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_0$ alloy, which is susceptible to pitting corrosion, has PRE value 16.87. This indicates that a PRE of above 26 is crucial to protect the as-cast Mo-containing alloys from pitting corrosion in 1 M NaCl solution.

The obtained polarisation results (Table 6) have shown that the Mo-containing alloys exhibit lower corrosion current density than that of the Mo-free alloy, which is in agreement with the other authors [50,58,77–78]. However, the corrosion current density of the Mo-containing alloys decreases as the Mo content increases from 0.1 to 0.8 mol. Although the PRE value for the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.8}$

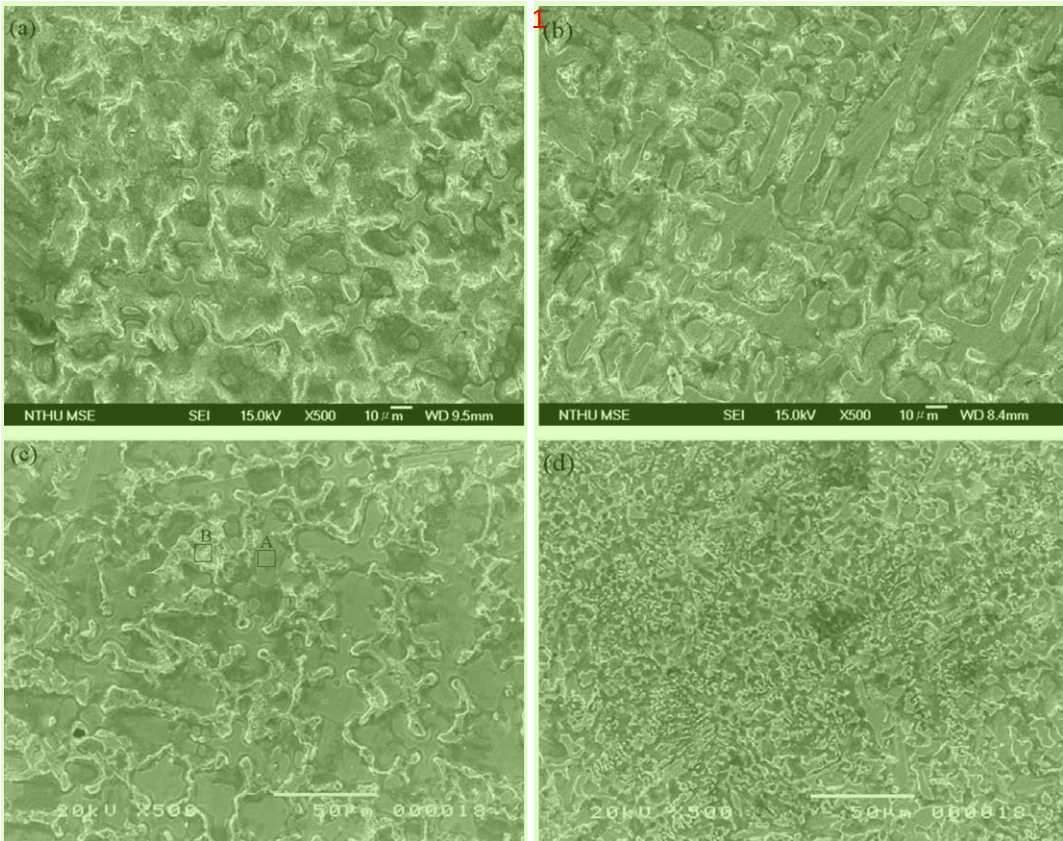


Fig. 6. SEM images of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ high-entropy alloys: (a) $x = 0$, (b) $x = 0.1$, (c) $x = 0.5$, and (d) $x = 0.8$ mol after potentiodynamic polarisation in deaerated 0.5 M H_2SO_4 .

Table 7
EDS results for the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.5}$ alloy with different selected area A and B after the potentiodynamic polarisation in deaerated 0.5 M H_2SO_4 .

Element	$\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.5}$ (atom%)	
	A	B
Co	21.9	24.4
Cr	20.7	18.0
Fe	12.8	17.6
Ni	13.8	27.1
Ti	6.9	6.5
Mo	23.9	6.4

is the highest among the Mo-containing alloys, the corrosion current density is the lowest among the Mo-containing alloys. The major drawbacks of the PRE equation in using parameters based on Cr and Mo contents is that they ignored the relationship between microstructure and pitting corrosion resistance. Kim et al. [83] investigated corrosion of σ phase formation of 25% Cr duplex stainless steel (DSS) in 4 M NaCl solutions. According to the author, the increase of Mo content in DSS has been reported to promote formation of σ phase, leading to a reduction in pitting corrosion resistance of the DSS. Similar results were found by Cvijović and Radenković [84] in studies of the microstructure and pitting corrosion resistance of the annealed DSS in 0.5 M NaCl solution. In this study, as the Mo content increases from 0.5 to 0.8 mol in the high-entropy alloys, the interdendrite phase of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys is a Cr and Mo rich σ phase, and the dendrite is a Cr and Mo depleted phase. Mo and Cr contents are higher in the interdendrite than in the dendrite, as can be observed in Table 2. The PRE value and the associated resistance to pitting may become notably different in the two phases. Therefore, the formation of σ phase in the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys can result in a significant degradation of pitting corrosion resistance in 1 M NaCl solutions.

$\text{Ni}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0.5$, and 0.8) alloys can result in a significant degradation of pitting corrosion resistance in 1 M NaCl solutions.

A passive film breaks down when the applied potential is raised into the transpassive region of the potentiodynamic polarisation curve, and the formation of pits can be initiated at discrete locations on the metal surface [85]. A passive film is generally believed to repair itself when the repassivation potential (E_{rp}) exceeds the corrosion potential. As listed in Table 6, the E_{rp} is more noble than the E_{corr} , regardless of whether Mo is incorporated. As the Mo content in the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys increases from 0 to 0.1 mol, the E_{rp} rises from 32.5 to 761 mV_{SHE}. However, as the Mo content in the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys increases from 0.1 to 0.8 mol, the E_{rp} falls from 761 to 476 mV_{SHE}. It is assumed that the presence of the σ phase in the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ ($x = 0.5, 0.8$) alloys reduces the repassivation performance. Therefore, the addition of appropriate Mo to $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys improves the ability to restore its passive film.

3.4. SEM photomicrographs of corroded surfaces

Fig. 6 presents the SEM microstructure of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys after potentiodynamic polarisation in deaerated 0.5 M H_2SO_4 . According to the SEM images, surface examination showed that the major types of corrosion were mainly uniform corrosion. As the Mo content increases from 0.5 to 0.8 mol in the high-entropy alloys, according to the Table 2, the interdendrite phase of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys is a Cr and Mo rich phase, and the dendrite is a Cr and Mo depleted phase. The active dendrites are coupled to the Cr and Mo depleted and Cr and Mo rich passive interdendrites. The low anode/cathode surface area ratio results in galvanic corrosion at the dendrites, and the

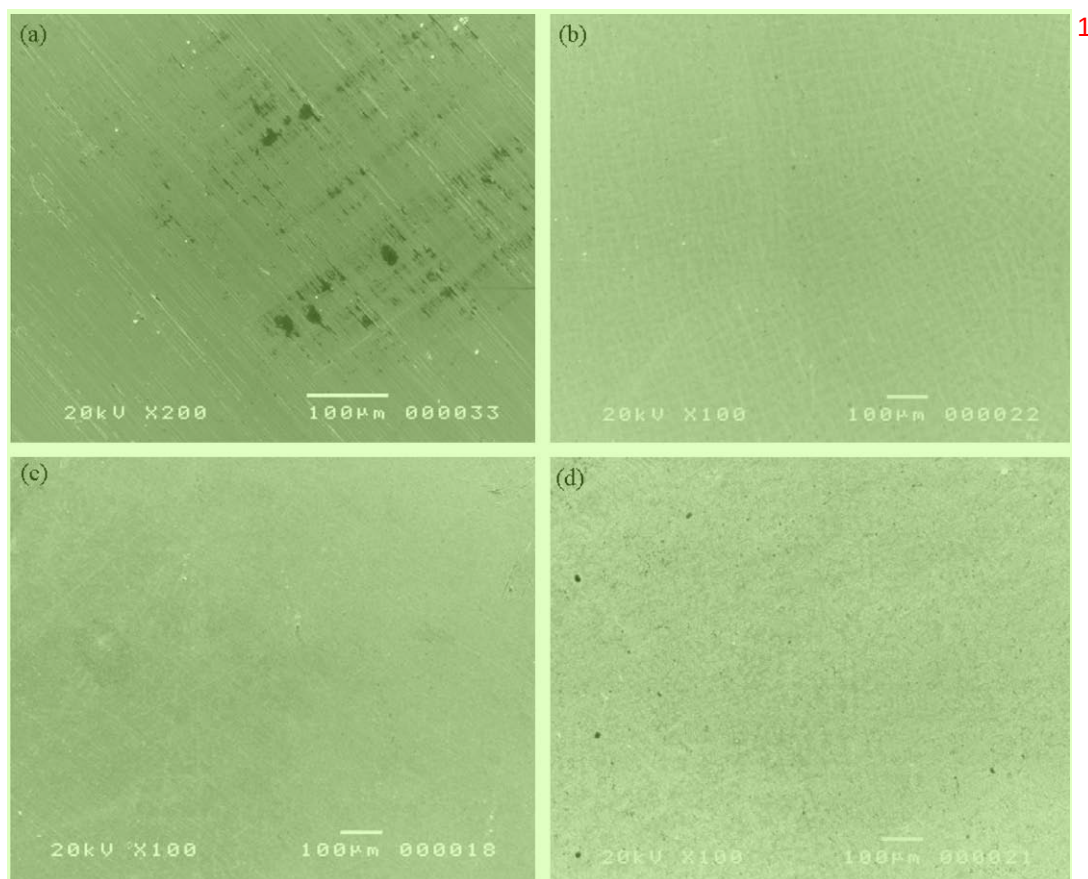


Fig. 7. SEM images of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ high-entropy alloys: (a) $x = 0$, (b) $x = 0.1$, (c) $x = 0.5$, and (d) $x = 0.8$ mol after potentiodynamic polarisation in deaerated 1 M NaCl.

electrochemical action causes attack along the dendrites in Cr and Mo depleted alloys. To verify the composition of the interdendrite phase, the EDS analysis was used, and the result is listed in Table 7. There are two areas, designated A and B in Fig. 6c after the potentiodynamic polarisation of the alloy $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.5}$ in deaerated 0.5 M H_2SO_4 . Area A is the interdendrite and area B is the dendrite. The EDS analysis demonstrates that the interdendrites of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.5}$ alloy are rich in Cr and Mo, but poor in Co, Ni, and Fe. Especially, the atomic percent of Mo in the interdendrites (23.9 atom%) considerably exceeds that in the dendrites (6.4 atom%) in the $\text{Co}_{1.5}\text{CrFeNi}_{0.5}\text{Ti}_{0.5}\text{Mo}_{0.5}$ alloy. Therefore, the corrosion on the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys ($x = 0.5$, and 0.8) alloys was concentrated to the dendrites.

The SEM micrographs of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys after potentiodynamic polarisation in deaerated 1 M NaCl are shown in Fig. 7. The corroded surfaces of the Mo-free alloy showed that the major types of corrosion were mainly pitting corrosion, not uniform corrosion. It is observed that several pits formed in 1 M NaCl on Mo-free alloy. Moreover, we find that there are no pits on the surface of Mo-containing alloys, supporting the previous finding that Mo-containing alloys are free of pitting corrosion in 1 M NaCl solution. It appears, therefore, that the Mo-free alloy is more susceptible to pitting corrosion in 1 M NaCl solution than the Mo-containing alloys. These results are in agreement with the cyclic polarisation curves as presented in Fig. 5.

Fig. 8a–d illustrates the SEM microstructure of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys after potentiodynamic polarisation in deaerated 1 M NaOH. It is clear that the corrosion mode in NaOH is more of general type, indicating passivating nature of the alloys in this oxidising solution. The pit-like contrasts on the $\text{Co}_{1.5}\text{CrFe}$

$\text{Ni}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys ($x = 0, 0.1, 0.5$, and 0.8) alloys are due to the microporosities formed during the casting and the mechanical polishing effect, compared to Fig. 2, images of the as-cast $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys with various Mo contents. The surface of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys ($x = 0.1, 0.5$, and 0.8) alloys is roughened by general corrosion. Similar results were found by Mondal et al. [22] in the studies of the electrochemical behaviour of multi-component amorphous and nanocrystalline zirconium (Zr)-based alloys in different environments, which demonstrated that pitting has occurred in the alloy in NaCl solutions whereas no pitting has observed in NaOH and H_2SO_4 solutions.

5. Conclusions

The electrochemical characteristics of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ high-entropy alloys in solutions of 0.5 M H_2SO_4 , 1 M NaCl, and NaOH have been studied in this paper. The potentiodynamic polarisation curves of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys in acidic solution exhibit active–passive corrosion behaviour, yielding an extensive passive region. The corrosion current density and passive current density of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}$ alloy are significantly lower than those of the $\text{Co}_{1.5}\text{CrFeNi}_{0.5}\text{Ti}_{1.5}\text{Mo}_{0.1}$, $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.5}$, and $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_{0.8}$ alloys. Therefore, the Mo-free alloy is more resistant to general corrosion than that of the Mo-containing alloys in acidic environments. On the other hand, the cyclic polarisation curves indicate that the Mo-free alloy tends to pitting corrosion in 1 M NaCl solution accounting for the lower breakdown potential and narrower range of potentials in the passive region. Additionally, the negative hysteresis in the reverse scans of the cyclic polar-

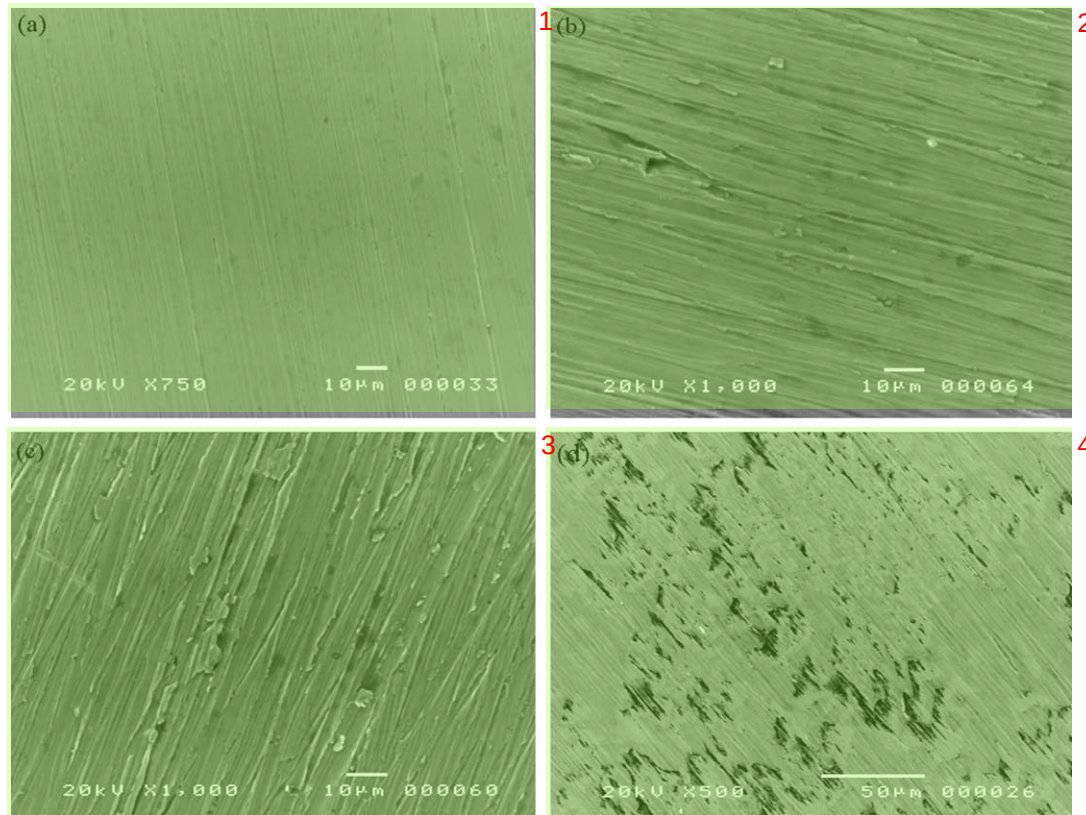


Fig. 8. SEM images of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ high-entropy alloys: (a) $x = 0$, (b) $x = 0.1$, (c) $x = 0.5$, and (d) $x = 0.8$ mol after potentiodynamic polarisation in deaerated 1 M NaOH.

isation curves and SEM images show that the Mo-containing alloys are not susceptible to pitting corrosion in deaerated 1 M NaCl solution in this study, and that the passive film is generally believed to repair itself. It is revealed that the addition of Mo in the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys improves the pitting resistance in chloride environments, in that the breakdown potential for the Mo-containing alloys are much higher than that for the Mo-free alloy. Potentiodynamic polarisation curves of the $\text{Co}_{1.5}\text{CrFeNi}_{1.5}\text{Ti}_{0.5}\text{Mo}_x$ alloys, obtained in 1 M NaOH solution, clearly reveal that the corrosion resistance decreases with the Mo content increases.

References

- [1] J.R. Davis (Ed.), Metals Handbook, 10th ed., vol. 1, ASM International, Metals Park, OH, 1990.
- [2] J.R. Davis (Ed.), Metals Handbook, 10th ed., vol. 2, ASM International, Metals Park, OH, 1990.
- [3] A. Inoue, T. Zhang, T. Masumoto, Glass-forming ability of alloys, *J. Non-Cryst. Solids* 156 (1993) 473–480.
- [4] A. Greer, Confusion by design, *Nature* 366 (1993) 303–304.
- [5] A. Peker, W.L. Johnson, A highly processable metallic glass: $\text{Zr}_{41.2}\text{Ti}_{13.8}\text{Cu}_{12.5}\text{Ni}_{10.0}\text{Be}_{22.5}$, *Appl. Phys. Lett.* 63 (1993) 2342–2344.
- [6] A. Inoue, K. Amiya, T. Zhang, T. Masumoto, Mechanical strength and thermal stability of Ti-based amorphous alloys with large glass-forming ability, *Mater. Sci. Eng. A* 179 (1994) 692–696.
- [7] T. Zhang, A. Inoue, New bulk glassy Ni-based alloys with high strength of 3000 MPa, *Mater. Trans. JIM* 43 (2002) 708–711.
- [8] K. Asami, S.J. Pang, T. Zhang, A. Inoue, Preparation and corrosion resistance of Fe–Cr–Mo–C–B–P bulk glassy alloys, *J. Electrochem. Soc.* 149 (2002) B366–B369.
- [9] A. Inoue, M. Koshiba, T. Itoi, A. Makino, Ferromagnetic Co–Fe–Zr–B amorphous alloys with glass transition and good high-frequency permeability, *Appl. Phys. Lett.* 73 (1998) 744–746.
- [10] A. Inoue, W. Zhang, T. Zhang, K. Kurosaka, High-strength Cu-based bulk glassy alloys in Cu–Zr–Ti and Cu–Hf–Ti ternary systems, *Acta Mater.* 49 (2000) 2645–2652.
- [11] B. Shen, M. Akiba, A. Inoue, Effect of Cr addition on the glass-forming ability, magnetic properties, and corrosion resistance in FeMoGaPCBSi bulk glassy alloys, *J. Appl. Phys.* 100 (2006) 043523.
- [12] S.J. Pang, T. Zhang, K. Asami, A. Inoue, Bulk glassy Fe–Cr–Mo–C–B alloys with high corrosion resistance, *Corros. Sci.* 44 (2002) 1847–1856.
- [13] I. Chatteraj, S. Baunack, M. Stoica, A. Gebert, Electrochemical response of $\text{Fe}_{65.5}\text{Cr}_4\text{Mo}_4\text{Ga}_4\text{P}_{12}\text{C}_5\text{B}_{5.5}$ bulk amorphous alloy in different aqueous media, *Mater. Corros.* 55 (2004) 36–42.
- [14] D.Y. Liu, H.F. Zhang, Z.Q. Hu, W. Gao, Magnetic and corrosion properties of $\text{Fe}_{56}\text{Co}_7\text{M}_2\text{Mo}_5\text{Zr}_{10}\text{B}_{20}$ ($M = \text{W}$ or Ni) bulk metallic glasses, *J. Alloys Compd.* 422 (2006) 28–31.
- [15] H. Habazaki, A. Kawashima, K. Asami, K. Hashimoto, The corrosion behaviour of amorphous Fe–Cr–Mo–P–C and Fe–Cr–W–P–C alloys in 6 M HCl solution, *Corros. Sci.* 33 (1992) 225–236.
- [16] M.W. Tan, E. Akiyama, A. Kawashima, K. Asami, K. Hashimoto, The effect of air exposure on the corrosion behaviour of amorphous Fe–8Cr–Mo–13P–7C alloys in 1 M HCl solution, *Corros. Sci.* 37 (1995) 1289–1301.
- [17] M.W. Tan, E. Akiyama, A. Kawashima, K. Asami, K. Hashimoto, The influence of Mo addition and air exposure on the corrosion behaviour of amorphous Fe–8Cr–13P–7C alloys in de-aerated 1 M HCl, *Corros. Sci.* 38 (1996) 349–365.
- [18] H. Katagiri, S. Meguro, M. Yamasaki, H. Habazaki, T. Sato, A. Kawashima, K. Asami, K. Hashimoto, Synergistic effect of three corrosion-resistant elements on corrosion resistance in concentrated hydrochloric acid, *Corros. Sci.* 43 (2001) 171–182.
- [19] H. Habazaki, T. Sato, A. Kawashima, K. Asami, K. Hashimoto, Preparation of corrosion-resistant amorphous Ni–Cr–P–B bulk alloys containing molybdenum and tantalum, *Mater. Sci. Eng. A* 304–306 (2001) 696–700.
- [20] M.L. Morrison, R.A. Buchanan, A. Peker, W.H. Peter, J.A. Horton, P.K. Liaw, Cyclic-anodic-polarisation studies of a $\text{Zr}_{41.2}\text{Ti}_{13.8}\text{Ni}_{10}\text{Cu}_{12.5}\text{Be}_{22.5}$, *Intermetallics* 12 (2004) 1177–1181.
- [21] A. Dhawan, S. Roychowdhury, P.K. De, S.K. Sharma, Potentiodynamic polarisation studies on bulk amorphous alloys and $\text{Zr}_{46.75}\text{Ti}_{8.25}\text{Cu}_{7.5}\text{Ni}_{10}\text{Be}_{27.5}$ and $\text{Zr}_{65}\text{Cu}_{17.5}\text{Ni}_{10}\text{Al}_{7.5}$, *J. Non-Cryst. Solids* 351 (2005) 951–955.
- [22] K. Mondal, B.S. Murty, U.K. Chatterjee, Electrochemical behaviour of multicomponent amorphous and nanocrystalline Zr-based alloys in different environments, *Corros. Sci.* 48 (2006) 2212–2225.
- [23] L. Liu, B. Liu, Influence of the micro-addition of Mo on glass forming ability and corrosion resistance of Cu-based bulk metallic glasses, *Electrochim. Acta* 51 (2006) 3724–3730.
- [24] L. Liu, B. Liu, Improvement of corrosion resistance of Cu-based bulk metallic glasses by the microalloying of Mo, *Intermetallics* 15 (2007) 679–682.

- [25] L. Liu, B. Liu, The effect of microalloying on thermal stability and corrosion resistance of Cu-based bulk metallic glasses, *Mater. Sci. Eng. A* 415 (2006) 286–290.
- [26] K. Asami, C.L. Qin, T. Zhang, A. Inoue, Effect of additional elements on the corrosion behaviour of a Cu–Zr–Ti bulk metallic glass, *Mater. Sci. Eng. A* 375–377 (2004) 235–239.
- [27] D. Zander, B. Heisterkamp, I. Gallino, Corrosion resistance of Cu–Zr–Al–Y and Zr–Cu–Ni–Al–Nb bulk metallic glasses, *J. Alloys Compd.* 434–435 (2007) 234–236.
- [28] S. Pang, C.H. Shek, C. Ma, A. Inoue, T. Zhang, Corrosion behaviour of a glassy Ti–Zr–Hf–Cu–Ni–Si alloy, *Mater. Sci. Eng. A* 449–451 (2007) 557–560.
- [29] J. Jayaraj, K.B. Kim, H.S. Ahn, E. Fleury, Corrosion mechanism of HCl-containing Fe–Cr–Mo–Y–C–B bulk amorphous alloys in highly concentrated HCl solution, *Mater. Sci. Eng. A* 449–451 (2007) 517–520.
- [30] A. Dhawan, K. Sachdev, S. Roychowdhury, P.K. De, S.K. Sharma, Potentiodynamic polarisation studies on amorphous $Zr_{46.75}Ti_{8.25}Cu_{7.5}Ni_{10}Be_{27.5}$, $Zr_{65}Cu_{17.5}Ni_{10}Al_{7.5}$, $Zr_{67}Ni_{33}$ and $Ti_{60}Ni_{40}$ in aqueous HNO_3 solutions, *J. Non-Cryst. Solids* 353 (2007) 2619–2623.
- [31] M.W. Tan, E. Akiyama, H. Habazaki, A. Kawashima, K. Asami, K. Hashimoto, The role of chromium and molybdenum in passivation of amorphous Fe–Cr–Mo–P–C alloys in de-aerated 1 M HCl, *Corros. Sci.* 38 (1996) 2137–2151.
- [32] H. Katagiri, S. Meguro, M. Yamasaki, H. Habazaki, T. Sato, A. Kawashima, K. Asami, K. Hashimoto, An attempt at preparation of corrosion-resistant bulk amorphous Ni–Cr–Ta–Mo–P–B alloys, *Corros. Sci.* 43 (2001) 183–191.
- [33] S. Hiromoto, A.P. Tsai, M. Sumita, T. Hanawa, Effect of chloride ion on the anodic polarisation behaviour of the $Zr_{65}Al_{7.5}Ni_{10}Cu_{17.5}$ amorphous alloy in phosphate buffered solution, *Corros. Sci.* 42 (2000) 1651–1660.
- [34] T.C. Chieh, J. Chu, C.T. Liu, J.K. Wu, Corrosion of $Zr_{52.5}Cu_{17.5}Ni_{4.6}Al_{10}Ti_5$ bulk metallic glasses in aqueous solutions, *Mater. Lett.* 57 (2003) 3022–3025.
- [35] J.W. Yeh, S.K. Chen, J.Y. Gan, S.J. Lin, T.S. Chin, T.T. Shun, C.H. Tsau, S.Y. Chang, Formation of simple crystal structure in Cu–Co–Ni–Cr–Al–Fe–Ti–V alloys with multiprincipal metallic elements, *Metall. Mater. Trans. A* 35A (2004) 2533–2536.
- [36] C.J. Tong, M.R. Chen, S.K. Chen, J.W. Yeh, T.T. Shun, S.J. Lin, S.Y. Chang, Mechanical performance of the $Al_{50}CoCrCuFeNi$ high-entropy alloy system with multiprincipal elements, *Metall. Mater. Trans. A* 36A (2005) 1263–1271.
- [37] C.Y. Hsu, J.W. Yeh, S.K. Chen, T.T. Shun, Wear resistance and high-temperature compression strength of fcc $CuCoNiCrAl_{0.5}Fe$ alloy with boron addition, *Metall. Mater. Trans. A* 35A (2004) 1465–1469.
- [38] P.K. Huang, J.W. Yeh, T.T. Shun, S.K. Chen, Multi-principal-element alloys with improved oxidation and wear resistance for thermal spray coating, *Adv. Eng. Mater.* 6 (2004) 74–78.
- [39] J.W. Yeh, S.K. Chen, S.J. Lin, J.Y. Gan, T.S. Chin, T.T. Shun, C.H. Tsau, S.Y. Chang, Nanostructured high-entropy alloys with multiple principal elements: novel alloy design concepts and outcomes, *Adv. Eng. Mater.* 6 (2004) 299–303.
- [40] R.A. Swalin, *Thermodynamics of Solids*, second ed., Wiley, New York, 1991.
- [41] Y.Y. Chen, T. Duval, U.D. Hung, J.W. Yeh, H.C. Shih, Microstructure and electrochemical properties of high entropy alloys—a comparison with type-304 stainless steel, *Corros. Sci.* 47 (2005) 2257–2279.
- [42] C.P. Lee, Y.Y. Chen, C.Y. Hsu, J.W. Yeh, H.C. Shih, The effect of boron on the corrosion resistance of the high entropy alloys $Al_{0.5}CoCrCuFeNi_{8.5}$, *J. Electrochem. Soc.* 154 (2007) C424–C430.
- [43] C.P. Lee, C.C. Chang, Y.Y. Chen, J.W. Yeh, H.C. Shih, Effect of the aluminium content of $Al_xCrFe_{1.5}MnNi_{0.5}$ high-entropy alloys on the corrosion behaviour in aqueous environments, *Corros. Sci.* 50 (2008) 2053–2060.
- [44] H.Y. Chen, C.W. Tsai, C.C. Tung, J.W. Yeh, T.T. Shun, C.C. Yang, S.K. Chen, Effect of the substitution of Co by Mn in Al–Cr–Cu–Fe–Co–Ni high-entropy alloys, *Ann. Chim. Sci. Mater.* 31 (2006) 685–698.
- [45] Y.C. Wang, J.W. Yeh, Microstructure and Mechanical Properties of $Al_xCo_{1.5}CrFeMo_yNi_{1.5}Ti_{0.5}$, Master's thesis of National Tsing Hua University, Taiwan, 2007.
- [46] Y.S. Huang, L. Chen, H.W. Lui, M.H. Cai, J.W. Yeh, Microstructure, hardness, resistivity and thermal stability of sputtered oxide films of $AlCoCrCu_{0.5}NiFe$ high-entropy alloy, *Mater. Sci. Eng. A* 457 (2007) 77–83.
- [47] T.K. Chen, T.T. Shun, J.W. Yeh, M.S. Wong, Nanostructured nitride films of multi-element high-entropy alloys by reactive DC sputtering, *Surf. Coat. Technol.* 188–189 (2004) 193–200.
- [48] H.F. Kuo, W. Chin, T.W. Cheng, W.K. Hsu, J.W. Yeh, Hyperfine splitting from magnetic boride domains embedded in Fe–Co–Ni–Al–B–Si alloy, *Appl. Phys. Lett.* 89 (2006) 182503.
- [49] D.A. Jones, *Principles and Prevention of Corrosion*, second ed., Prentice Hall, New Jersey, 1996.
- [50] M.A. Ameer, A.M. Fekry, F. El-Taib Heikal, Electrochemical behaviour of passive films on molybdenum-containing austenitic stainless steels in aqueous solutions, *Electrochim. Acta* 50 (2004) 43–49.
- [51] Y.Y. Chen, L.B. Chou, H.C. Shih, Factors affecting the electrochemical behaviour and stress corrosion cracking of Alloy 690 in chloride environments, *Mater. Chem. Phys.* 97 (2006) 37–49.
- [52] B.D. Cullity, *Elements of X-ray Diffraction*, Addison-Wesley, Massachusetts, 1978.
- [53] A. Takeuchi, A. Inoue, Calculations of mixing enthalpy and mismatch entropy for ternary amorphous alloys, *Mater. Trans.* 41 (2000) 1372–1378.
- [54] A. Takeuchi, A. Inoue, Calculations of amorphous-forming composition range for ternary alloy systems and analyses of stabilisation of amorphous phase and amorphous-forming ability, *Mater. Trans.* 42 (2001) 1435.
- [55] R. Boom, A.R. Miedema, A.K. Niessen, F.R. de Boer, W.C. Mattens, *Cohesion in Metals: Transition Metal Alloys*, Elsevier Applied Science, Amsterdam, The Netherlands, 1988.
- [56] H.S. Khatak, B. Raj, *Corrosion of Austenitic Stainless Steels: Mechanism Mitigation and Monitoring*, Woodhead Publishing, 2002.
- [57] O. Conejero, M. Palacios, S. Rivera, Premature corrosion failure of a 316L stainless steel plate due to the presence of sigma phase, *Eng. Fail. Anal.* 16 (2009) 699–704.
- [58] A.J. Sedriks, Effects of alloy composition and microstructure on the passivity of stainless steels, *Corrosion* 42 (1986) 376–389.
- [59] T. Laitinen, M. Bojinov, I. Betova, K. Mäkelä, T. Saario, The properties of and Transport Phenomena in Oxide Films on Iron, Nickel, Chromium and their Alloys in Aqueous Environments, STUK-YTO-TR150. ISBN 951-712-286-1, Helsinki, January 1999.
- [60] K. Sugimoto, Y. Sawada, The role of molybdenum additions to austenitic stainless steels in the inhibition of pitting in acid chloride solutions, *Corros. Sci.* 17 (1977) 425–445.
- [61] K. Hashimoto, K. Asami, K. Teramoto, An X-ray photo-electron spectroscopic study on the role of molybdenum in increasing the corrosion resistance of ferritic stainless steels in HCl, *Corros. Sci.* 19 (1979) 3–14.
- [62] H. Ogawa, H. Omata, I. Itoh, H. Okada, Auger electron spectroscopic and electrochemical analysis of the effect of alloying elements on the passivation behaviour of stainless steels, *Corrosion* 34 (1978) 52–60.
- [63] Y.C. Lu, C.R. Clayton, Evidence for a bipolar mechanism of passivity in Mo bearing stainless steels, *J. Electrochem. Soc.* 132 (1985) 2517–2518.
- [64] A.R. Brooks, C.R. Clayton, K. Doss, Y.C. Lu, On the role of Cr in the passivity of stainless steel, *J. Electrochem. Soc.* 133 (1986) 2459–2464.
- [65] C.R. Clayton, Y.C. Lu, A bipolar model of the passivity of stainless steel: the role of Mo addition, *J. Electrochem. Soc.* 133 (1986) 2465–2473.
- [66] T. Kodama, J.R. Ambrose, Effect of molybdate ion on the repassivation kinetics of iron in solutions containing chloride ions, *Corrosion* 33 (1977) 155–161.
- [67] K. Ogura, T. Ohama, Pit formation in the cathodic polarisation of passive iron. IV: Repair mechanism by molybdate, chromate and tungstate, *Corrosion* 40 (1984) 47–51.
- [68] M. Pourbaix, *Atlas of electrochemical equilibria in aqueous solutions*, NACE, Houston, TX, 1974.
- [69] K. Hashimoto, K. Asami, A. Kawashima, H. Habazaki, E. Akiyama, The role of corrosion-resistant alloying elements in passivity, *Corros. Sci.* 49 (2007) 42–52.
- [70] M. Bojinov, G. Fabricius, T. Laitinen, K. Mäkelä, T. Saario, G. Sundholm, Influence of molybdenum on the conduction mechanism in passive films on iron–chromium alloys in sulphuric acid solution, *Electrochim. Acta* 46 (2001) 1339–1358.
- [71] J.O. Nilsson, T. Huhtala, P. Jonsson, L. Karlsson, A. Wilson, Structural stability of super duplex stainless weld metals and its dependence on tungsten and copper, *Metall. Mater. Trans. A* 27A (1996) 2196–2208.
- [72] Y. Maehara, Y. Ohmori, J. Murayama, N. Fujino, T. Kunitake, Effects of alloying elements on sigma phase precipitation in delta-gamma duplex phase stainless steels, *Metal. Sci.* 17 (1983) 541–547.
- [73] D.M.E. Villanueva, F.C.P. Junior, R.L. Plaut, A.F. Padilha, Comparative study on sigma phase precipitation of three types of stainless steels: austenitic, superferritic and duplex, *Mater. Sci. Technol.* 22 (2006) 1098–1104.
- [74] H.H. Uhlig, R.W. Revie, *Corrosion and Corrosion Control: An Introduction to Corrosion Science and Engineering*, third ed., Wiley, New York, 1985.
- [75] W.S. Tait, *An Introduction to Electrochemical Corrosion Testing for Practicing Engineers and Scientists*, Racine, Wisconsin, 1994.
- [76] B.E. Wilde, A critical appraisal of some popular laboratory electrochemical tests for predicting the localized corrosion resistance of stainless alloys in sea water, *Corrosion* 28 (1972) 283–291.
- [77] R.J. Brigham, Pitting of molybdenum bearing austenitic stainless steel, *Corrosion* 28 (1972) 177–179.
- [78] K. Sugimoto, Y. Sawada, The role of alloyed molybdenum in austenitic stainless steels in the inhibition of pitting in neutral halide solutions, *Corrosion* 32 (1976) 347–352.
- [79] J.D. Bumgardner, L.C. Lucas, Surface analysis of nickel chromium dental alloys, *Dent. Mater.* 9 (1993) 252–259.
- [80] M. Roach, D. Parsell, S. Gardner, J.D. Bumgardner, Correlation of corrosion and surface analyses for Ni–Cr alloys, *Crit. Rev. Biomed. Eng.* 26 (1998) 391–392.
- [81] M.B. Rockel, M. Renner, Pitting crevice and stress corrosion resistance of high chromium and molybdenum alloy stainless steels, *Mater. Corros.* 35 (1984) 537–542.
- [82] T.J. Glover, Recent developments in corrosion-resistant metallic alloys for construction of seawater pumps, *Mater. Perform.* 27 (1988) 51–56.
- [83] J.S. Kim, H.S. Kwon, Effects of Tungsten on corrosion and kinetics of sigma phase formation of 25% chromium duplex stainless steels, *Corrosion* 55 (1999) 512–521.
- [84] Z. Cvijović, G. Radenković, Microstructure and pitting corrosion resistance of annealed duplex stainless steel, *Corros. Sci.* 48 (2006) 3887–3906.
- [85] Z. Szklarska-Smialowska, *Pitting Corrosion of Metals*, NACE, Houston, Texas, 1986.